Skills Summary

Multiplying and Dividing Decimals by Place Value

Use this reference while completing your worksheet or when studying.

Skill 1 — Multiply or divide by 10, 100, 1000

Rule: the digits move, the decimal point stays put.

$\times 10$ moves every digit **one place left** (each digit is worth ten times more); $\times 100$ moves it two places left. $\div 10$ moves every digit **one place right**; $\div 100$ moves it two places right. Fill empty places with 0.

Example: 3.6×100
Step 1. Multiplying by 100 makes each digit worth 100 times as much, so every digit slides two places to the left.
Step 2. The 3 ones become 3 hundreds and the 6 tenths become 6 tens: **360**.

Try it: $47.5 \div 100$

check: **0.475**

Skill 2 — Multiply two decimals

Rule: multiply the digits as whole numbers, then place the point by place value.

tenths $\times$ tenths $=$ hundredths and hundredths $\times$ tenths $=$ thousandths.

Shortcut: the product has as many decimal places as the two factors have together.

Example: 0.6×0.4
Step 1. Both factors are tenths, so multiply the digits first and expect an answer in hundredths.
Step 2. $6 \times 4 = 24$, and 24 hundredths is **0.24**.

Try it: 1.2×0.5

check: **0.6**

Skill 3 — Divide by a decimal

Rule: rewrite both numbers in the **same place-value unit**, then divide whole numbers.

$3.5 \div 0.7$ becomes "how many 7 tenths fit in 35 tenths?"

Dividing by a number smaller than 1 gives a quotient *larger* than the number you started with.

Example: $3.5 \div 0.7$
Step 1. Both numbers can be said in tenths, and counting how many 7 tenths fit inside 35 tenths turns this into whole-number division.
Step 2. $35 \div 7 = 5$, and the check $5 \times 0.7 = 3.5$ works, so the quotient is **5**.

Try it: $5.6 \div 0.8$

check: **7**

Skill 4 — Estimate first, then judge the size

Rule: round each number to the nearest whole, multiply or divide those, and use the result to test where the decimal point belongs.

Multiplying by a number *smaller than one* makes the result smaller; dividing by a number *smaller than one* makes the result larger.

Example: Estimate 5.8×3.1

Step 1. An estimate only needs whole numbers, so round each factor to the nearest whole before multiplying.

Step 2. 5.8 rounds to 6 and 3.1 rounds to 3, giving $6 \times 3 = \mathbf{18}$.

Try it: Estimate 7.2×4.9

check: **35**

(!) Watch out: A product of two numbers below 1 must be smaller than both of them. If your answer is bigger, the decimal point moved the wrong way.